

Ecesis Investments
2026 Summer Power Systems Modeling Internship

Assignment 1: Constraint Mapping Across Data Sources

*Building a Translation Layer Between Three Dialects
of the PJM Grid Language*

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github.com/GavinGongTech/ECESIS-2026-Summer-Power-Systems-Modeling-Assignment

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Abstract

This report documents a three-layer constraint mapping pipeline that reconciles PJM Market, Dayzer (Cambridge Energy Solutions/PowerGEM), and Panorama (Enverus) transmission constraint datasets. The same physical transmission constraint is represented differently across these three systems due to fundamental differences in their naming conventions and data origins. The pipeline combines string normalization, domain-informed heuristics grounded in PJM documentation, fuzzy string matching, and sentence-transformer ML matching to produce a unified output with honest confidence scoring. Of 5,230 Market constraints, 73.3% were matched to at least one external source with confidence ≥ 0.85 . Manual validation confirms 92.5% accuracy for exact matches and 85.0% for fuzzy matches. Key limitations and future improvements are documented with full transparency.

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Introduction

In power markets, the same physical transmission constraint rarely carries a single name. PJM’s market constraint list, the Dayzer FTR trading platform, and the Enverus Panorama congestion analytics database each describe PJM’s transmission network using their own naming conventions, formats, and levels of granularity. For a trading desk like Ecesis Investments, this creates a critical operational problem: without a reliable mapping between these three sources, it is impossible to form a coherent view of congestion risk across data providers.

This assignment tasks us with building a “translation layer” between three dialects of the same grid language. Each source defines constraints as (facility, contingency) pairs, but the naming conventions differ substantially:

- **Market (PJM)** uses EMS B3 naming conventions as documented in PJM Manual 03A Section 2.2.4 [2] — human-readable facility names with voltage levels and abbreviated endpoints.
- **Dayzer (CES/PowerGEM)** derives names from PSS/E bus data processed through the TRANZER database [4, 5], producing compact facility identifiers with underscore separators and different abbreviation schemes.
- **Panorama (Enverus)** uses PJM’s EMS naming convention but stores historical binding constraint occurrences spanning 16 years (2010–2026) [8, 9, 10].

The core challenge is that these differences are not arbitrary formatting variations — they reflect fundamentally different data pipelines, each with its own internal logic. A naive string comparison approach will fail. A robust solution requires understanding what each source is, where its data comes from, and why its naming conventions differ from the others.

Approach Overview

The pipeline implements three sequential matching layers, applied in order of confidence:

1. **Exact matching** after string normalization — highest confidence
2. **Fuzzy string matching** using RapidFuzz — medium-high confidence
3. **ML semantic matching** using sentence transformers — medium confidence

Additionally, specialized matching handles transfer interfaces (via PJM Manual 03 Section 3.8 canonical lookup [1]) and Limiting Outage (L/O) entries (via parsed facility-outage decomposition). Every output row carries a confidence score, match type, and flags so the trading desk knows exactly which mappings to trust.

Data Description

Source Identification and Provenance

A thorough literature review was conducted to identify the origin and purpose of each dataset before any matching was attempted. The findings are summarized in Table 1.

Table 1: Data source identification and key properties

Property	Market (PJM)	Dayzer (CES)	Panorama (Enverus)
Rows	5,230	13,813	21,963
Columns	6	2	5
Source	PJM official market data	CES/PowerGEM TRANZER	Enverus Panorama
Naming convention	EMS B3 (PJM M-3A §2.2.4)	PSS/E bus names	EMS B3 (mixed)
Scope	Currently monitored	FTR trading model	Historical binding events
Coverage	Current snapshot	Current model	2010–2026 (16.3 years)
Confidence	~99% (authoritative)	~87% (inferred)	~95%

PJM Market Dataset

The Market dataset is PJM’s authoritative list of currently monitored transmission constraints. Each row represents a unique (CONSTRAINT, CONTINGENCY) pair where CONSTRAINT is the monitored facility and CONTINGENCY is the simulated outage event [1]. The REPORTEDNAME column contains the EMS B3 display name, which is the primary matching anchor against external sources.

Key structural properties: 5,230 rows with zero duplicates on the (CONSTRAINT, CONTINGENCY) key; 2,710 unique facilities; 1,070 unique contingencies; contingency types dominated by L500 (21.4%), Base Case/ACTUAL (24.2%), and L230 (13.3%).

Dayzer Dataset

Dayzer is a product of Cambridge Energy Solutions (CES), now owned by PowerGEM [4, 7]. It was designed for FTR trading and daily PJM data synchronization [5, 6]. The data pipeline is: PJM publishes PSS/E network files → TRANZER imports raw PSS/E data into an Access database → BusMap maps network topology → NAPD database → Dayzer NAME column [4].

The NAME column encodes (facility:contingency) as a single colon-separated string. Of 13,813 rows: 12,912 (93.5%) use one colon, 262 (1.9%) use two colons (compound contingencies), and 639 (4.6%) use no colon (interface/special entries). The larger row count relative to Market reflects Dayzer’s broader FTR model scope.

Panorama Dataset

Panorama is Enverus’s high-performance powerflow engine for congestion analytics, explicitly covering ERCOT, SPP, MISO, and PJM [8, 9]. Based on Enverus documentation and empirical analysis, the dataset is interpreted as a historical binding constraint occurrence database: each row records a unique (Monitored Facility, Contingency Name) pair along with the earliest and latest timestamps of that constraint appearing as binding [10].

Key structural properties: 21,963 rows; sorted by Latest timestamp descending (confirmed monotonic decreasing); timestamps spanning January 2010 to April 2026 (16.3 years). The Monitored Facility

column uses two formats: 39.7% parenthetical (“SUBST A kV – SUBST B kV (REPORTEDNAME)”) and 60.3% non-parenthetical.

Important caveat: No official Enverus data dictionary was found confirming column definitions. The interpretation of Earliest/Latest as first and most recent binding occurrence is inferred from data structure and is consistent with Enverus’s documented capability to “map years of market data to state estimator data” [10].

Why Panorama Has the Most Rows

Market tracks currently monitored constraints (~5,230); Dayzer covers FTR model scope (~13,774 after cleaning); Panorama accumulates every unique (facility, contingency) pair that has ever appeared as binding over 16 years (~21,884 after cleaning). Constraints that are no longer active, have been renamed, or bound only historically contribute to Panorama’s larger count without appearing in Market’s current list.

Cross-Source Contingency Overlap

A key empirical finding from EDA: 87.2% of Market contingencies appear exactly in Dayzer after case normalization, and 99.5% appear in Panorama. This confirms that contingency names are highly standardized across sources and serve as the primary matching anchor, despite facility names being more variable.

Methodology

Data Cleaning and Normalization

All text fields were normalized using the following pipeline:

1. Uppercase conversion
2. Strip leading/trailing whitespace
3. Collapse multiple internal spaces to single space
4. Normalize voltage notation: 138KV → 138 KV
5. Base case normalization: ACTUAL / BASE / Actual → __BASE_CASE__ sentinel (PJM Manual 03 Section 1.3 [1])
6. SCTNLZ variant normalization: (Sctnlz) → (SCTNLZ) (PJM Manual 03 Attachment E [1])

Data quality issues resolved before matching:

- **Dayzer:** 3 placeholder entries dropped (JK, ABC, 5018); 36 Type B duplicate NAMES deduplicated (keep highest CID)
- **Panorama:** 79 normalized duplicate (MF, CN) pairs deduplicated (keep highest PID)
- **Market:** 4 entries sharing the same normalized display name handled via MATCH_KEY incorporating CONSTRAINTID

Domain Heuristic Functions

Four domain-informed functions enforce physical constraints on the matching process:

Voltage level hard filter: If two facility strings reference different valid PJM voltage levels (69, 115, 138, 161, 230, 345, 500, or 765 kV — confirmed from PJM Manual 03 Section 3.3 [1]), confidence

is set to 0.0 regardless of string similarity. A 138 kV and 345 kV line at the same substation are physically different elements.

Directional normalization: Facility names of the form “SUBSTATION_A – SUBSTATION_B” are normalized by sorting endpoints alphabetically, resolving asymmetric naming across sources.

Circuit number agreement: For structured contingencies in PJM format (*Lk V.Sub1-Sub2.circuit*), circuit numbers are extracted and compared. A mismatch applies a 0.70 confidence multiplier, reflecting that the same two substations can have multiple circuits.

L/O parsing: Panorama Limiting Outage entries (format: “FACILITY L/O OUTAGE”) are split into two components and matched separately.

Confidence Scoring Framework

Every matched row receives a composite confidence score:

$$\text{confidence} = \text{base_score} \times m_{\text{method}} \times m_{\text{circuit}} \quad (1)$$

$$\text{base_score} = 0.6 \times \text{facility_score} + 0.4 \times \text{contingency_score} \quad (2)$$

where facility and contingency scores are string similarity values in $[0, 1]$. The 60/40 weighting reflects that facility names encode the physical element identity (substation name, voltage level, line abbreviation) while contingency names provide disambiguation. Sensitivity analysis confirmed results are stable under 50/50 and 70/30 weighting.

Method multipliers:

Table 2: Confidence method multipliers

Method	Multiplier	Score range
Exact match	1.00	1.00
Fuzzy match	0.95	0.85–0.94
ML (sentence transformer)	0.90	0.75–0.88
Interface lookup	0.90	0.90
Partial L/O match	0.75	0.70–0.75

Confidence tiers: High (≥ 0.95), Medium-High (0.85–0.95), Medium (0.70–0.85), Low (0.50–0.70), Unmatched (< 0.50).

For three-way matches, overall confidence is $\min(\text{dayzer_confidence}, \text{pano_confidence})$ — the most conservative approach, treating a three-way match as only as strong as its weakest link.

Matching Pipeline

Layer 1: Exact Matching

Pre-built lookup indexes (Python dictionaries) allow $O(1)$ exact match lookup for each Market row against Dayzer and Panorama. Market REPORTEDNAME is matched first; CONSTRAINT is used as fallback.

For Panorama, two strategies were tested in parallel:

- **Strategy A:** Match Market REPORTEDNAME against Panorama inside-parentheses content
- **Strategy B:** Match Market REPORTEDNAME against Panorama outside-parentheses content

Strategy A yielded 1,376 exact matches vs. Strategy B's 435 — confirming that Panorama's parenthetical content corresponds to Market's REPORTEDNAME format (resolving Gap 1, consistent with PJM Manual 03A Section 2.2.4 [2]).

Layer 2: Fuzzy Matching

RapidFuzz `token_sort_ratio` is applied to facility names (handles word order differences) and `ratio` to structured contingency names. Threshold cross-validation tested values from 0.70 to 0.95 on a 200-row sample. The selected threshold of 0.85 was chosen conservatively; threshold sensitivity analysis confirms the curve plateaus below 0.85, dropping by only 159 matches from 0.85 to 0.70.

Layer 3: ML Matching (Sentence Transformers)

Applied only to entries unresolved by Layers 1 and 2, specifically targeting descriptive text contingencies that share no character-level similarity with Market's structured format codes. The `all-MiniLM-L6-v2` model [13] was selected for its training on semantic similarity tasks, computational efficiency, and status as the community standard for general-purpose sentence similarity. Embeddings are L2-normalized; cosine similarity serves as the match score. Threshold: 0.75.

Important caveat: The model was trained on general English text, not power systems terminology. Substation abbreviations and circuit codes are not well-represented in its training data, limiting effectiveness for highly technical entries.

Special Category Matching

Transfer interface constraints (Eastern, Western, APSOUTH, BED-BLA, AEP-DOM, CE-EAST, CLEVELAND, BC/PEPCO) are matched using a hardcoded canonical lookup table derived from PJM Manual 03 Section 3.8 [1]. L/O entries are parsed and matched at lower confidence (0.75 method multiplier), with unmatched entries exported for manual review.

Results

Overall Coverage

Table 3 summarizes the final matching results across all layers.

Table 3: Final matching results by match type

Match Type	Count	% of Market	Description
Three-way	508	9.7%	All three sources agree
Market + Panorama	3,148	60.2%	Bilateral Panorama match
Market + Dayzer	162	3.1%	Bilateral Dayzer match
Interface	60	1.1%	Transfer interface lookup
Market + Pano (L/O)	1	0.0%	Limiting outage match
Dayzer + Panorama	9	—	Non-Market bilateral
Market unmatched	1,412	27.0%	No confident match found
Total output rows	5,300		
Overall Market coverage	3,879	73.3%	

Table 4: Match method contribution

Method	Count	%	Confidence tier
Exact	1,822	34.4%	High (≥ 0.95)
Fuzzy	1,826	34.5%	Medium-High (0.85–0.95)
ML	177	3.3%	Medium (0.75–0.88)
Interface lookup	60	1.1%	Medium-High (0.90)
Partial L/O	1	0.0%	Medium
Unmatched	1,412	26.6%	—

Confidence Distribution

Of 3,888 matched rows: 1,795 (33.9%) achieve high confidence (≥ 0.95); 1,934 (36.5%) achieve medium-high confidence (0.85–0.95); 159 (3.0%) achieve medium confidence (0.70–0.85). No matches were accepted below the 0.70 threshold. The confidence distribution shows two distinct clusters: exact matches near 1.0 and fuzzy matches tightly clustered in the 0.85–0.90 band, consistent with a well-calibrated matching pipeline.

Source-Level Coverage

Table 5: Coverage by source

Source	Total entries	Matched	Coverage	Primary method
Market (PJM)	5,230	3,879	73.3%	Exact + Fuzzy
Panorama (Enverus)	21,884	3,726	17.0%	Exact + Fuzzy
Dayzer (CES)	13,774	738	5.4%	Fuzzy

The low Panorama and Dayzer coverage from Market’s perspective is expected and informative: both sources contain far more constraints than Market because they cover broader scope (FTR model and 16-year history respectively). 95% of Dayzer entries and 84% of Panorama entries have no current Market counterpart.

Validation

Four independent validation checks were performed:

Manual spot-check (30 samples): Stratified across match methods, each sample assessed on a four-point scale (CORRECT / LIKELY_CORRECT / UNCERTAIN / WRONG). Results: exact 92.5%, fuzzy 85.0%, ML 70.0%, overall 82.5%. ML matches performed at the lower end of expectations consistent with the identified contingency type disagreements (see Section 5.2).

Voltage agreement rate: 100% for Market-Dayzer pairs; 99.6% for Market-Panorama pairs (the 0.4% apparent mismatch reflects transformer entries where Panorama names both voltage sides — not true matching errors).

Contingency type agreement: 99.9% agreement on structured contingency kV levels across all Market-Panorama matches.

Threshold sensitivity: The match count curve shows a clear cliff at 0.90 (1,741 matches removed), confirming that fuzzy matches cluster tightly in the 0.85–0.90 band. The 0.85 threshold is well-positioned at the natural boundary.

Discussion

Key Analytical Findings

Finding 1 — Panorama’s parenthetical content corresponds to Market’s REPORTED-NAME format. Strategy A (inside parentheses) yielded $3.2\times$ more exact matches than Strategy B (outside parentheses): 1,376 vs. 435. This confirms that Enverus stores the EMS B3 display name inside parentheses when adopting the full “SUBST A – SUBST B (REPORTEDNAME)” format, consistent with PJM Manual 03A Section 2.2.4 [2].

Finding 2 — Unmatched rows are primarily a facility naming gap, not a scope gap. Of 1,412 unmatched Market rows, 1,135 (80.4%) have their contingency present in Panorama — the facility name failed to match, not the contingency. This is a critical distinction: it means these constraints are matchable in principle, and improving facility normalization is the highest-priority improvement for a future iteration. Visual inspection of unmatched base case entries (e.g. LINVFT_230KVLIN-VFT1_1_LN) confirms that Dayzer-style underscore separators in some Market constraint names prevent matching.

Finding 3 — 138 kV constraints are the most unmatched voltage level (31.5%). Lower-voltage constraints bind less frequently historically and therefore have fewer Panorama entries, contributing disproportionately to the unmatched count.

Finding 4 — Dayzer coverage is low due to structural naming differences. At 5.4% coverage, Dayzer’s PSS/E-derived bus names differ structurally from Market’s EMS B3 names in ways that fuzzy matching only partially bridges. This is expected given the different data pipelines and is not a pipeline failure.

Finding 5 — L/O manual review would recover ~432 additional matches. Manual inspection of 10 sampled L/O entries found 6 (60%) with facility names present in Market data. Extrapolating: ~432 of 720 L/O entries are potentially recoverable through systematic manual review.

Limitations

Dayzer naming conventions: No official CES/PowerGEM documentation was found confirming Dayzer’s naming convention. The colon separator interpretation (facility:contingency) is based on 87.2% empirical contingency overlap with Market — strong but not officially confirmed.

Panorama column definitions: No official Enverus data dictionary was found defining PID, Monitored Facility, Contingency Name, Earliest, or Latest. All interpretations are inferred from data structure and Enverus product documentation.

ML match quality: 76 of 143 Panorama ML matches (53%) show contingency type disagreement — the ML model matched facility names semantically but the contingency types differ. These 158 medium-confidence ML matches should be verified before use in trading decisions.

Three identified wrong matches: Three rows match Market contingency L115.MIDDLETOWNJCT-COLLINS.975 (SCTNLZ) to Panorama contingency L138.MiddletownJct-Collins (Sctnlz) — different voltage levels in the contingency name. The voltage hard filter applies to facility names only; contingency voltage level checking is a noted gap.

Raw EMS entries: 610 Panorama contingency entries containing CB OPEN, DUMMY, and CMPTRAF sequences are unmatchable without access to PJM’s EMS model.

L/O automated matching: Automated L/O matching yielded only 1 confident match despite 720 successfully parsed entries, due to format asymmetry with Market’s single-element constraint structure.

Three-way confidence conservatism: Using $\min(\text{dayzer}, \text{pano})$ for three-way confidence is conservative — a Panorama exact match paired with Dayzer fuzzy yields overall confidence 0.91 despite the Panorama component being certain.

Future Work

1. **Priority 1 — Underscore normalization:** Add `s.replace('_', ' ')` to `normalize_facility()`. Section 9 analysis suggests this single change could recover a substantial portion of the 1,135 facility naming gap cases.
2. **Priority 2 — L/O manual review:** Systematic review of `lo_unmatched_for_review.csv` estimated to recover ~432 additional matches.
3. **Priority 3 — ML match validation:** Manual verification of 158 medium-confidence ML matches; contingency voltage level checking.
4. **Priority 4 — Separate confidence reporting:** Report Dayzer and Panorama confidence separately for three-way matches rather than taking the minimum.
5. **Priority 5 — CES/Enverus documentation:** Inquiry submitted to PowerGEM (AZobian@CES-US.com) regarding Dayzer naming conventions. No response received prior to submission.

Conclusion

This report presents a three-layer constraint mapping pipeline that reconciles PJM Market, Dayzer, and Panorama transmission constraint datasets. Of 5,230 Market constraints, 73.3% were matched to at least one external source with confidence ≥ 0.85 , validated at 82.5% overall accuracy by manual spot-check.

The 26.7% unmatched rate is intentional — the pipeline reports NULL rather than forcing low-confidence mappings, because an incorrect match presented with false confidence is more dangerous to a trading desk than an honest acknowledgment of uncertainty. The root cause analysis in Section 9 shows that 80.4% of unmatched rows are addressable through facility normalization improvements, while 19.4% represent genuine scope gaps that no matching method can resolve without additional data.

The primary differentiator of this pipeline is its grounding in domain knowledge: naming convention differences are not treated as arbitrary formatting noise but as consequences of specific data engineering choices documented in PJM manuals, CES product documentation, and Enverus product materials. This understanding directly informs the matching strategy and is reflected in the citations throughout this report.

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